

LANGCHAIN2

RAG REST API

Source: README.md

🖥️ Project : Deploy Application as a Web Service with Render

server deploying an OpenAI chat model and a chain to tell a joke about a topic.

🔧 Requirements : PYTHON Installation & Setup

Python 3.11

```
brew install pyenv pyenv install 3.11.11
```

switch to Python 3.11

```
pyenv local 3.11.11
```

🌐 Create a virtual environment & activate the virtual environment :

MacOS/Linux:

```
python3 -m venv env  
source env/bin/activate
```

Windows:

```
python -m venv env  
.\env\Scripts\activate
```

🔧 Installation:

Install Python 3.11

use pip3 on a Mac or Linux and pip on Windows

```
pip install -r requirements.txt
```

Get an API key

Set OpenAI API Key as an environment variable:

```
export OPENAI_API_KEY='sk-brHeh...A39v5iXsM2'
```

.env file:

```
OPENAI_API_KEY=sk-brHeh...A39v5iXsM2
```

Start the server:

```
uvicorn main:app --reload
```

Test the RESTAPI

joke/invoke

```
curl --location --request POST 'http://localhost:8000/joke/invoke' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "input": {
      "topic": "cats"
    }
  }'
```

joke/batch

```
curl --location --request POST 'http://localhost:8000/joke/batch' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "inputs": [
      {"topic": "cats"},
      {"topic": "dogs"}
    ]
  }'
```

Try the Playground & Docs :

<http://localhost:8000/joke/playground/> <http://localhost:8000/openai/playground/>

Display a documentation

<http://localhost:8000/docs>

Deploy Live to Render

For Render, make sure the service root is [24.LangChain2/Project6/RAG-REST-API](#) and use:

- Build command: `pip install -r requirements.txt`
- Start command: `uvicorn main:app --host 0.0.0.0 --port $PORT`

This project includes a `.python-version` file that pins Render to Python `3.11`, which avoids `tiktoken==0.6.0` being built from source on Render's default Python `3.14`.

Do not install additional deployment packages outside `requirements.txt`. Keeping Render on the pinned dependency set prevents resolver drift and avoids incompatibilities such as `langserve==0.2.2` with newer `httpx` releases.